

AkaiRyu | RedDragon | ■■

OPERATION AKAIRYU (■■■)

Threat Actor	MirrorFace (Earth Kasha) — APT10 Subgroup
Campaign Name	Operation AkaiRy■ (Japanese: RedDragon)
Timeline	June – September 2024
Classification	TLP: WHITE — Public Research
Report Date	May 12, 2026
Primary Source	ESET Research (Dominik Breitenbacher, March 2025)
MITRE Group ID	G1054

Prepared for: Cybersecurity Research & Threat Intelligence Community
References: ESET WeLiveSecurity | MITRE ATT&CK® | SOC Prime | The Hacker News

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1. EXECUTIVE SUMMARY

Operation AkaiRyū (黒龍 — Japanese for **RedDragon**) is a sophisticated, multi-stage cyber-espionage campaign attributed to **MirrorFace** (also tracked as **Earth Kasha**), a China-aligned advanced persistent threat (APT) group assessed with high confidence to be a subgroup of the APT10 umbrella — officially designated **menuPass** in MITRE ATT&CK® (Group G0045).

First uncovered by ESET Research in Q2–Q3 2024, AkaiRyū represents a landmark operational shift: for the first time ever, MirrorFace extended its targeting beyond Japanese entities to strike a **Central European diplomatic institute**. The campaign introduced a refreshed toolset including the **ANEL backdoor** (believed abandoned since 2018–2019), a heavily customised **AsyncRAT** variant running inside **Windows Sandbox**, and abuse of **VS Code Remote Tunnels** for covert command-and-control.

Key findings from this report: the threat actor leveraged Expo 2025 (Osaka) as a social-engineering lure, abused signed legitimate applications (McAfee, JustSystems) for DLL side-loading, and maintained operational security by running malware inside Windows Sandbox to evade host-based detection. MirrorFace is formally catalogued in MITRE ATT&CK® under **Group G1054**.

SEVERITY	SCOPE	ATTRIBUTION CONFIDENCE	PRIMARY MOTIVATION
CRITICAL	Diplomatic, Academia, Defense, Media	HIGH — China-aligned (APT10 / menuPass)	Cyber Espionage / Data Exfiltration

2. CAMPAIGN OVERVIEW

Operation AkaiRy■ was active between **June and September 2024**. The campaign was publicly disclosed by ESET Research on **18 March 2025** in a detailed blog post on WeLiveSecurity. The operation is aligned with what Japan's National Police Agency (NPA) and the National Center of Incident Readiness and Strategy for Cybersecurity (NCSC) refer to internally as **Campaign C**.

The campaign name 'AkaiRy■' was coined by ESET to reflect the campaign's geographic theme (Japan) and the threat actor's China-state nexus. The name translates directly as *RedDragon* — evoking both the Chinese national symbol and the persistent threat posed by PRC-aligned actors.

Campaign Highlights

- **First European Target:** A Central European diplomatic institute — the first confirmed MirrorFace attack outside Japan.
- **Expo 2025 Lure:** Spearphishing emails referencing the World Expo 2025 in Osaka were used to gain the victim's trust.
- **ANEL Resurrection:** The ANEL (UPPERCUT) backdoor — dormant since 2018/2019 — was revived as the group's primary first-stage backdoor.
- **AsyncRAT in Windows Sandbox:** A heavily customised AsyncRAT variant was deployed inside Windows Sandbox, bypassing most host-based security controls.
- **VS Code Abuse:** Visual Studio Code's Remote Tunnels feature was leveraged for covert, encrypted command-and-control.
- **Signed Binary Abuse:** McAfee and JustSystems signed executables were weaponised through DLL side-loading to evade detection.
- **Credential Theft:** Google Chrome user data — including saved passwords, contact information, and credit card details — was exfiltrated.

3. THREAT ACTOR / GROUP ASSOCIATION

Identity & Attribution

Attribute	Detail
Primary Name	MirrorFace
Aliases	Earth Kasha, APT10-MirrorFace subgroup
MITRE ATT&CK® ID	G1054
Parent Group	menuPass (APT10) — G0045
Nation-State Nexus	People's Republic of China (PRC)
Active Since	At least 2019
Primary Language	Mandarin Chinese (assessed)
Motivation	State-sponsored cyber espionage, intelligence collection
Flagship Backdoor	HiddenFace (NOOPDOOR)
Historical Backdoor	LODEINFO (2022–2023), ANEL/UPPERCUT (revived 2024)

APT10 Umbrella Connection

MirrorFace's use of the **ANEL backdoor** — previously considered exclusive to APT10 (menuPass) — is the strongest technical indicator linking the group to the broader APT10 umbrella. ESET, Trend Micro, and MITRE ATT&CK® all assess MirrorFace as a subgroup operating under the menuPass umbrella (G0045), sharing tooling, infrastructure patterns, and targeting philosophy.

APT10 (menuPass) has been active since at least 2006, attributed to the Chinese Ministry of State Security (MSS), and is known for targeting managed service providers, government entities, aerospace, defence, and financial organisations globally. MirrorFace inherits this targeting philosophy while focusing primarily on Japan-related entities.

Historical Operations by MirrorFace

Operation	Year	Primary Target	Key Tool
LiberalFace	2022	Japanese Political Entities	LODEINFO
MirrorFace Campaigns	2022–2023	Japanese Media, Defence, Finance	LODEINFO, HiddenFace
AkaiRyu (Campaign C)	Jun–Sep 2024	Japan + Central European Diplomatic Institute	ANEL, AsyncRAT, HiddenFace

4. CAMPAIGN TIMELINE

Date	Event	Target
Jun 20, 2024	Spearphishing attack via malicious password-protected Word document with VBA macro (mouseover trigger). ANEL v5.5.4 loaded via signed McAfee executable.	Japanese Research Institute (2 employees)
Aug 26, 2024	Initial benign reconnaissance email sent referencing Expo 2025 in Osaka — leveraging a prior legitimate interaction with a Japanese NGO.	Central European Diplomatic Institute
Aug 2024 (Follow-up)	Second email containing a malicious OneDrive link to a ZIP archive with a disguised LNK file. LNK triggers PowerShell chain, deploys ANEL v5.5.5 via JustSystems-signed binary (DLL side-loading).	Central European Diplomatic Institute
Post-Compromise Aug–Sep 2024	Deployment of HiddenFace backdoor, AsyncRAT (in Windows Sandbox), VS Code Remote Tunnels, PuTTY, Rubeus (Kerberos interaction). Chrome credential/data exfiltration confirmed on one machine.	Central European Diplomatic Institute
Jan 2025	Japan NPA/NCSC publish public warning regarding MirrorFace 'Campaign C' targeting academia, think tanks, politicians, and media in Japan.	Public Advisory
Mar 18, 2025	ESET publicly discloses Operation AkaiRyu findings in WeLiveSecurity blog post. Results also presented at JSAC 2025.	Public Disclosure
Apr–May 2025	MITRE ATT&CK;® v19 formally adds MirrorFace (G1054) as a subgroup of menuPass (G0045), with Operation AkaiRyu as a named campaign.	MITRE ATT&CK;® Database

5. OBJECTIVES OF ATTACK

Operation AkaiRy■ is a **cyber espionage** campaign with multiple interrelated intelligence-collection objectives. The attack was not financially motivated. Based on ESET's forensic findings and MirrorFace's historical targeting patterns, the following objectives are assessed with high confidence:

Objective 1: Intelligence Collection on Japan–EU Relations

The Central European diplomatic institute targeted maintains active diplomatic relationships and communication channels with Japanese government bodies and NGOs. The Expo 2025 lure demonstrates specific awareness of these relationships. Intercepting diplomatic communications would directly benefit Chinese intelligence.

Objective 2: Credential and Identity Theft

Confirmed exfiltration of Google Chrome data including stored passwords, autofill data, contact lists, and credit card information. These credentials may enable further intrusions into related networks or facilitate social engineering.

Objective 3: Long-Term Persistent Access

Deployment of multiple backdoors (ANEL, HiddenFace) and VS Code Remote Tunnels indicates intent to maintain persistent, long-term covert access — consistent with an ongoing intelligence collection mission rather than a smash-and-grab.

Objective 4: Network Reconnaissance and Lateral Movement

Tools such as Rubeus (Kerberos attack framework) and PuTTY were deployed on a secondary machine to facilitate lateral movement and deeper network enumeration.

Objective 5: Expanding Geographic Reach

This operation marks a deliberate strategic expansion beyond Japan, suggesting that Chinese intelligence tasking has broadened to include European diplomatic entities with Japan connections — potentially in preparation for Expo 2025 or related geopolitical events.

6. TARGETED SECTORS & COUNTRIES

Sectors Targeted (2019–2024)

Sector	Relevance in AkaiRyu	Historical Priority
Diplomacy / Government	Primary target (EU diplomatic institute)	CRITICAL
Academia / Think Tanks	Japanese research institutes targeted	HIGH
Defence / Aerospace	Historical MirrorFace targeting	HIGH
Media / Journalism	NPA warning — Campaign C overlap	HIGH
Finance / Banking	Chrome credential theft implications	MEDIUM
Manufacturing	Historical APT10 targeting	MEDIUM
Politics / Elections	2022 Japanese election spearphishing	HIGH

Geographic Targeting

Country/Region	Role in Campaign	Status
Japan	Primary historical focus; research institute targeted Jun 2024	CONFIRMED
Central Europe (EU)	Diplomatic institute targeted Aug 2024 — first European target	CONFIRMED
Taiwan	Trend Micro reported MirrorFace activity in Taiwan (related campaigns)	REPORTED
India	Trend Micro / Vietnamese NCSC reported cases	REPORTED
Vietnam	Reported activity overlapping MirrorFace operations	REPORTED

7. ATTACK FLOW

The attack flow below describes the **Central European diplomatic institute** intrusion in August 2024 — the most thoroughly documented incident in Operation AkaiRyū, reconstructed through ESET's collaborative forensic investigation.

Phase 1 — Reconnaissance & Initial Contact

MirrorFace operators conducted prior open-source intelligence (OSINT) on the diplomatic institute, identifying a previous legitimate interaction between the institute and a Japanese NGO. This was used to craft a highly contextual, convincing spearphishing pretext referencing the World Expo 2025 in Osaka.

Phase 2 — Initial Access (Spearphishing)

Step 1: A benign first email was sent to the target referencing Expo 2025 and the prior NGO interaction — to establish trust and elicit a reply. (ATT&CK: T1566.002 — *Spearphishing Link*)

Step 2: After the target responded, a follow-up email was sent containing a **malicious OneDrive link** pointing to a ZIP archive. The archive contained a disguised **LNK (shortcut) file**. (ATT&CK: T1566.002, T1204.001 — *User Execution: Malicious Link*)

Phase 3 — Execution & DLL Side-Loading

Step 3: When the victim opened the LNK file, a chain of PowerShell commands was triggered, dropping additional malicious files to disk. (ATT&CK: T1059.001 — *PowerShell*)

Step 4: A **JustSystems-signed legitimate application** was used to side-load and decrypt the **ANEL backdoor v5.5.5** — a DLL side-loading technique that abuses trusted signed binaries to bypass application whitelisting and AV detection. (ATT&CK: T1574.002 — *DLL Side-Loading*)

Phase 4 — Persistence & Backdoor Deployment

Step 5: ANEL established initial backdoor access. MirrorFace subsequently deployed **HiddenFace (NOOPDOOR)** — their flagship persistent backdoor — to ensure long-term resilience. (ATT&CK: T1547 — *Boot/Logon Autostart*, T1543 — *Create/Modify System Process*)

Step 6: **VS Code Remote Tunnels** were deployed for encrypted, low-profile C2 communication — tunnelling through Microsoft's legitimate infrastructure to avoid network-based detection. (ATT&CK: T1572 — *Protocol Tunneling*, T1071 — *Application Layer Protocol*)

Phase 5 — Evasion via Windows Sandbox

Step 7: A heavily customised **AsyncRAT** variant was deployed using a multi-stage execution chain. A scheduled task at machine startup launched Windows Sandbox with a custom configuration file. Inside the sandbox, a batch script extracted AsyncRAT from a password-protected 7-Zip archive and ran it, with a secondary scheduled task executing it every hour.

Files delivered for the Windows Sandbox AsyncRAT chain:

- 7z.exe / 7z.dll — Legitimate 7-Zip archiver binaries
- setup.exe (disguised) — Password-protected archive containing AsyncRAT
- Batch script (.bat) — Unpacks and launches AsyncRAT
- Windows Sandbox config (.wsb) — Defines networking, directory mapping, memory, startup batch

(ATT&CK: T1564.006 — *Hide Artifacts: Run Virtual Instance*, T1053.005 — *Scheduled Task*)

Phase 6 — Lateral Movement & Credential Theft

Step 8: On a second compromised machine, MirrorFace deployed **Rubeus** (a Kerberos attack/enumeration toolkit) and **PuTTY** for lateral movement and deeper network access. (*ATT&CK: T1550.003 — Pass the Ticket, T1021 — Remote Services*)

Step 9: On one machine, the threat actor exported **Google Chrome web data** including stored contact information, autofill data (keywords), and **saved credit card details**. (*ATT&CK: T1555.003 — Credentials from Web Browsers*)

Attack Chain Summary Diagram

[1] RECON	OSINT — prior NGO interaction identified
[2] INITIAL ACCESS	Spearphishing email with OneDrive link (LNK in ZIP)
[3] EXECUTION	LNK → PowerShell chain → JustSystems DLL side-load
[4] ANEL v5.5.5	First-stage backdoor deployed via signed binary abuse
[5] PERSISTENCE	HiddenFace backdoor + Scheduled Tasks installed
[6] C2 EVASION	VS Code Remote Tunnels (encrypted via Microsoft infra)
[7] SANDBOX RAT	AsyncRAT deployed inside Windows Sandbox (hourly persistence)
[8] LAT. MOVEMENT	Rubeus (Kerberos), PuTTY deployed on second machine
[9] EXFILTRATION	Chrome credentials, contacts, autofill, credit card data

8. MITRE ATT&CK® TECHNIQUES USED

All techniques below are drawn from official MITRE ATT&CK® Enterprise framework references. MirrorFace is catalogued under **Group G1054**. Technique IDs are verified against ATT&CK® v19.

Tactic	Technique ID	Technique Name	Implementation in AkaiRyu
Initial Access	T1566.001	Spearphishing Attachment	Password-protected Word doc with VBA macro (June 2024 — Japan target)
Initial Access	T1566.002	Spearphishing Link	Malicious OneDrive link to ZIP/LNK (Aug 2024 — EU diplomatic target)
Execution	T1059.001	Command & Scripting: PowerShell	LNK file triggers PowerShell command chain to drop payloads
Execution	T1059.003	Command & Scripting: Windows Command Shell	Batch scripts used within Windows Sandbox to execute AsyncRAT
Execution	T1204.001	User Execution: Malicious Link	Victim clicks OneDrive link, opens LNK file initiating infection
Execution	T1204.002	User Execution: Malicious File	Malicious Word document opened, triggering VBA execution on mouseover
Persistence	T1053.005	Scheduled Task/Job: Scheduled Task	Scheduled task runs AsyncRAT (via sandbox) at startup + hourly
Persistence	T1543.003	Create/Modify System Process: Windows Service	HiddenFace backdoor maintains persistent service-based execution
Priv. Escalation	T1055	Process Injection	ANEL and HiddenFace use process injection techniques for execution
Priv. Escalation	T1548.002	Abuse Elevation: Bypass UAC	Hidden Start tool used to bypass UAC for elevated execution
Defence Evasion	T1574.002	Hijack Execution Flow: DLL Side-Loading	McAfee and JustSystems signed binaries abused to side-load ANEL DLL
Defence Evasion	T1564.006	Hide Artifacts: Run Virtual Instance	AsyncRAT executed inside Windows Sandbox — isolated from host AV
Defence Evasion	T1140	Deobfuscate/Decode Files or Information	ANEL payload encrypted; decrypted in-memory at runtime
Defence Evasion	T1036.005	Masquerading: Match Legitimate Name	setup.exe used as disguise for AsyncRAT archive payload
Defence Evasion	T1070.004	Indicator Removal: File Deletion	Delivered tools deleted post-execution to hinder forensic investigation
Credential Access	T1555.003	Credentials from Web Browsers	Google Chrome web data exfiltrated: passwords, contacts, credit cards
Credential Access	T1558	Steal/Forge Kerberos Tickets	Rubeus deployed for Kerberos ticket abuse and lateral movement prep
Discovery	T1082	System Information Discovery	Standard post-compromise enumeration of host environment

Discovery	T1083	File and Directory Discovery	File enumeration performed during post-compromise activity
Lateral Movement	T1021	Remote Services	PuTTY deployed for SSH-based lateral movement to second machine
Lateral Movement	T1550.003	Use Alternate Auth: Pass the Ticket	Rubeus facilitates Kerberos pass-the-ticket lateral movement
Command & Control	T1071	Application Layer Protocol	VS Code Remote Tunnels use HTTPS/WSS — blend with legitimate traffic
Command & Control	T1572	Protocol Tunneling	VS Code tunnels route C2 through Microsoft's cloud infrastructure
Command & Control	T1219	Remote Access Software	AsyncRAT, VS Code Remote Tunnels used for ongoing remote access
Command & Control	T1102	Web Service	OneDrive used as malware delivery platform; VS Code as C2 relay
Exfiltration	T1041	Exfiltration Over C2 Channel	Data exfiltrated over established ANEL/AsyncRAT C2 channels
Exfiltration	T1560	Archive Collected Data	7-Zip used to archive payloads; likely used for data staging too

9. REAL-WORLD INCIDENTS

Incident 1 — Japanese Research Institute (June 2024)

On **20 June 2024**, MirrorFace targeted **two employees** at a Japanese research institute. The attack vector was a **malicious, password-protected Word document** delivered in an unconfirmed manner (email presumed). The document contained VBA macro code triggered by a **mouseover event** — an unusual trigger that evades macro-execution warnings and sandboxes that don't simulate mouse movement. The macro executed code to load ANEL backdoor **version 5.5.4** via a signed McAfee executable through DLL side-loading.

This incident is notable for the first confirmed use of the resurrected ANEL backdoor by MirrorFace — a tool not seen in the wild since 2018/2019.

Incident 2 — Central European Diplomatic Institute (August 2024)

In **late August 2024**, MirrorFace executed a two-stage spearphishing operation against a **Central European diplomatic institute**. The attackers first sent a **benign email** referencing Expo 2025 — using knowledge of a prior legitimate interaction between the institute and a Japanese NGO. After the target responded, a second email delivered a **malicious OneDrive link** containing a ZIP archive with a disguised LNK shortcut file.

Opening the LNK triggered a PowerShell execution chain that ultimately side-loaded **ANEL v5.5.5** via a JustSystems-signed binary. ESET performed a close forensic collaboration with the affected institute, confirming post-compromise deployment of HiddenFace, AsyncRAT (in Windows Sandbox), VS Code Remote Tunnels, PuTTY, and Rubeus. On one machine, **Google Chrome data was confirmed exfiltrated** including contact info, autofill data, and saved credit card details.

This marks the first confirmed MirrorFace attack against a European entity, disclosed publicly by ESET at JSAC 2025 and via WeLiveSecurity on 18 March 2025.

Corroborating Intelligence

Operation AkaiRy■ overlaps with **Campaign C** as documented by Japan's NPA and NCSC in their January 2025 advisory, which warned of MirrorFace targeting Japanese academia, think tanks, politicians, and media since June 2024. Trend Micro independently published overlapping findings in November 2024, confirming the return of ANEL and expanded targeting. MITRE ATT&CK® v19 (released April 2025) formally integrated these findings, adding MirrorFace as a named subgroup of menuPass with AkaiRy■ as a campaign.

10. DETECTION OPPORTUNITIES

Endpoint Detection

Windows Sandbox Execution of Suspicious Processes: Alert on processes spawned by WindowsSandbox.exe or wsb configuration files that launch batch scripts, 7-Zip, or unknown executables. Baseline legitimate Windows Sandbox usage in your environment.

Scheduled Task Creation by Non-Admin Processes: Monitor schtasks.exe or Task Scheduler API calls originating from non-administrative user contexts or scripting engines (PowerShell, cmd).

DLL Side-Loading from Unusual Paths: Alert when signed binaries (McAfee, JustSystems, or any trusted vendor) load DLLs from user-writable paths or temp directories (T1574.002 detection via Sysmon Event ID 7).

PowerShell Execution via LNK Files: Correlate ShellExecute events where LNK files spawn PowerShell with encoded commands (Event ID 4688 / Sysmon EID 1 + 3).

Chrome Web Data Access by Non-Browser Processes: Alert on file access to AppData\Local\Google\Chrome\User Data\Default\Web Data or Login Data by processes other than chrome.exe.

Network Detection

VS Code Remote Tunnel Connections: Monitor outbound connections to *.vscode-cdn.net, *.tunnels.api.visualstudio.com. Flag VS Code processes on servers or workstations where development tools are not expected.

Unexpected OneDrive / SharePoint Download Chains: Detect LNK or ZIP files downloaded from OneDrive links followed immediately by PowerShell spawning — a hallmark two-stage delivery.

Anomalous SMB / Kerberos Traffic: Rubeus activity generates anomalous Kerberos AS-REQ/TGS-REQ patterns. Enable Kerberos audit logging and alert on Pass-the-Ticket indicators (Event ID 4768, 4769 with unusual encryption types).

SIEM / Behavioural Analytics Rules

```
Rule: Process = 'powershell.exe' AND Parent = '*.lnk handler' AND CommandLine CONTAINS '-enc'
```

```
Rule: Image LOAD WHERE SignedBinary IN ('McAfee*', 'JustSystems*') AND DLL_Path CONTAINS '\\AppData\\'
```

```
Rule: Process = 'WindowsSandboxClient.exe' AND ChildProcess CONTAINS '7z.exe OR cmd.exe'
```

```
Rule: FileAccess WHERE TargetFile LIKE '%Chrome%Web Data%' AND Process != 'chrome.exe'
```

```
Rule: ScheduledTask CREATE WHERE Author = 'SYSTEM' AND Action CONTAINS 'WindowsSandbox'
```

Threat Hunting Queries (Conceptual)

- Hunt: Identify all LNK file executions in the past 30 days that spawned PowerShell or cmd as a child process.
- Hunt: Enumerate all scheduled tasks containing references to .wsb (Windows Sandbox configuration) files.
- Hunt: Search for 7z.exe executions with password-protected archive creation or extraction in unusual directories (AppData, Temp).
- Hunt: Identify all processes making outbound connections to *.tunnels.api.visualstudio.com or *.vscode-cdn.net on non-developer systems.

11. DEFENSIVE RECOMMENDATIONS

REC-01: Restrict Windows Sandbox Usage

Disable Windows Sandbox via Group Policy (GPO) on all workstations where it is not a business requirement. Sandbox is enabled through Windows Features and requires administrative privileges — audit which systems have it enabled. *MITRE Mitigation: M1038 — Execution Prevention*

REC-02: Block or Monitor VS Code Remote Tunnels

Implement network policy controls to block outbound connections to VS Code tunnel endpoints on sensitive systems. Where VS Code is required, whitelist specific developer endpoints and alert on any unexpected tunnel establishment. *MITRE Mitigation: M1031 — Network Intrusion Prevention*

REC-03: Macro Policy and Protected View Enforcement

Enforce Group Policy to disable VBA macros in Office documents received from the internet (Mark of the Web). Enable Protected View. Deploy Attack Surface Reduction (ASR) rules — specifically rules targeting Office macro execution and child process spawning. *MITRE Mitigation: M1040 — Behaviour Prevention on Endpoint*

REC-04: Application Whitelisting / WDAC

Deploy Windows Defender Application Control (WDAC) or AppLocker to prevent execution of unsigned or unexpected binaries from user-writable locations. Pay special attention to DLLs loaded by trusted signed applications. *MITRE Mitigation: M1038 — Execution Prevention*

REC-05: Email Security — Sandboxed Link Analysis

Deploy advanced email security solutions that dereference and sandbox URLs at click-time, not just at delivery. OneDrive links used as delivery vectors often bypass static URL reputation checks. *MITRE Mitigation: M1017 — User Training, M1021 — Restrict Web-Based Content*

REC-06: Credential Manager Hardening

Prohibit browsers from storing sensitive credentials on high-value workstations and servers. Consider deploying a PAM (Privileged Access Management) solution. Clear saved passwords from browser profiles on sensitive machines. *MITRE Mitigation: M1027 — Password Policies*

REC-07: Kerberos and Lateral Movement Hardening

Enable Protected Users Security Group for privileged accounts to prevent Kerberos ticket caching attacks (Rubeus). Monitor for AS-REP Roasting (accounts with 'Do not require Kerberos pre-authentication'). Implement tiered administration to limit lateral movement paths. *MITRE Mitigation: M1015 — Active Directory Configuration*

REC-08: Spearphishing Awareness Training

Conduct regular phishing simulation exercises that include contextual lures (e.g., Expo 2025, conference invites, partner NGO communications). Train staff to verify unexpected document links via out-of-band channels. *MITRE Mitigation: M1017 — User Training*

REC-09: Threat Intelligence Integration

Subscribe to ESET WeLiveSecurity, MITRE ATT&CK® updates, and JPCERT/CC advisories for timely MirrorFace/Earth Kasha IOC updates. Integrate IOCs from ESET's AkaiRy research into SIEM, EDR, and firewall block lists. Leverage SOC Prime's ATT&CK-mapped detection content for rapid deployment.

REC-10: Incident Response Readiness

Establish runbooks for APT-level intrusions including DLL side-loading, Windows Sandbox abuse, and VS Code tunnel detection. Conduct tabletop exercises simulating MirrorFace-style multi-stage campaigns targeting

diplomatic and research personnel.

12. INDICATORS OF COMPROMISE (IOCS)

The following IOC categories have been reported in relation to Operation AkaiRyu. Full hash lists and IP/domain IOCs are published by ESET in their official research disclosure (WeLiveSecurity, March 2025). Refer to the ESET GitHub repository for complete, up-to-date IOC files.

IOC Type	Description	Source
Malware Family	ANEL (UPPERCUT) v5.5.4, v5.5.5	ESET Research
Malware Family	HiddenFace (NOOPDOOR) — MirrorFace flagship backdoor	ESET / Trend Micro
Malware Family	Customised AsyncRAT variant (Windows Sandbox execution)	ESET Research
Loader	ANEL Loader (Anelldr) — side-loaded via signed binaries	ESET Research
Loader	FaceXInjector — HiddenFace loader component	ESET Research
Tool	Rubeus — open-source Kerberos attack framework	ESET Forensics
Tool	Hidden Start — UAC bypass utility	ESET Research
Technique	Windows Sandbox .wsb config files — suspicious content	ESET Research
Delivery	OneDrive-hosted ZIP containing disguised LNK shortcut	ESET Research
Delivery	Password-protected Word .docx with mouseover VBA trigger	ESET Research
Signed Binary	McAfee application (specific version — see ESET report)	ESET Research
Signed Binary	JustSystems application (specific version — see ESET report)	ESET Research
Lure Theme	World Expo 2025, Osaka, Japan — diplomatic/NGO context	ESET Research

Note: For full file hashes (SHA-256), specific network IOCs (C2 IPs/domains), and YARA rules, refer to ESET's official release at: <https://github.com/eset/malware-ioc/tree/master/mirrorface> and the WeLiveSecurity blog post dated 18 March 2025.

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END OF REPORT — OPERATION AKAIRYU CAMPAIGN RESEARCH

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